

# CUSTOMER NEXT PURCHASE PREDICTION SYSTEM

Machine Learning Approach for Predicting Customer Purchase Categories and  
Generating Personalized Recommendations

| MACHINE LEARNING PROJECT |                  |              |
|--------------------------|------------------|--------------|
| Python                   | Pandas / NumPy   | Scikit-learn |
| XGBoost                  | Machine Learning | Streamlit    |

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Portfolio-ready project documentation based on the supplied Jupyter Notebook and deployment application.

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# 1. Project Overview

The Customer Next Purchase Prediction & Personalized Recommendation System predicts the product category a customer is most likely to purchase next using historical purchasing behavior.

The project combines customer behavior analysis, exploratory data analysis, feature engineering, multi-class classification, Top-K prediction, personalized product recommendations, and an interactive Streamlit application.

**Main Goal: Predict a customer's next purchase category and use the prediction to recommend relevant products.**

## End-to-End Workflow

|                       |                              |                  |   |                 |
|-----------------------|------------------------------|------------------|---|-----------------|
| Customer Data         | →                            | Data Cleaning    | → | EDA             |
| Feature Engineering   | →                            | Target Creation  | → | ML Models       |
| Model Evaluation      | →                            | Top-K Prediction | → | Recommendations |
| Streamlit Application | Interactive Customer Insight |                  |   |                 |

The notebook explicitly creates next\_purchase\_category from chronological transaction history rather than using it as a raw input field. This is intended to avoid target leakage.

## 2. Business Problem & Objective

E-commerce businesses have large amounts of customer transaction history but may not know which product category a customer is likely to purchase next.

A predictive system can transform historical behavior into a ranked list of likely next-purchase categories and then connect that prediction to product recommendations.

### Problem Statement

**Use historical customer, product, and transaction data to predict the next purchase category for a customer.**

### Project Objectives

- Analyze customer purchasing behavior.
- Clean and prepare customer, product, and transaction data.
- Create the next-purchase target chronologically.
- Engineer customer-level behavioral features.
- Compare multiple classification models.
- Evaluate Top-1, Top-3, and Top-5 prediction performance.
- Generate personalized product recommendations.
- Deploy the workflow through Streamlit.

### Business Flow

**Customer History → Behavior Features → ML Prediction → Product Recommendation**

### 3. Dataset Overview

The notebook loads three relational datasets from an Excel workbook: Customers, Products, and Transactions.

| Dataset      | Rows   | Columns | Purpose                                    |
|--------------|--------|---------|--------------------------------------------|
| Customers    | 10,000 | 7       | Customer profile and segment information   |
| Products     | 500    | 8       | Product, category, brand, price and rating |
| Transactions | 55,000 | 12      | Historical purchase activity               |

#### Customer Segments

| Segment    | Count |
|------------|-------|
| Regular    | 3,790 |
| Budget     | 2,187 |
| Occasional | 1,682 |
| Premium    | 1,538 |
| High Value | 803   |

#### Product Categories

| Category           | Products |
|--------------------|----------|
| Electronics        | 77       |
| Grocery            | 67       |
| Fashion            | 51       |
| Mobile Accessories | 44       |
| Home & Kitchen     | 41       |
| Footwear           | 36       |
| Beauty             | 34       |
| Sports             | 34       |
| Computers          | 27       |
| Books              | 25       |
| Home Appliances    | 18       |
| Office Supplies    | 15       |
| Personal Care      | 13       |
| Bags               | 11       |
| Gaming             | 7        |

## 4. Data Preparation & Target Creation

### Data Cleaning

The notebook converts date fields to datetime, removes exact duplicate rows, fills missing product ratings using category medians, and fills missing delivery days using the transaction median.

### Missing Values

| Dataset      | Notable Missing Values | Treatment              |
|--------------|------------------------|------------------------|
| Customers    | None reported          | No imputation required |
| Products     | 22 product ratings     | Category median        |
| Transactions | 498 delivery days      | Overall median         |

### Creating the Target

The raw data does not contain the target. The notebook creates next\_purchase\_category by shifting each customer's chronological category sequence by one transaction.

Purchase 1 → Category A   then   Purchase 2 → Category B  
Target for Purchase 1 = Category B

Rows without a known next purchase are excluded from the final modeling dataset. The resulting modeling workflow uses customer behavioral features to predict the next category.

### Train / Test Split

| Setting        | Notebook Value |
|----------------|----------------|
| Test size      | 20%            |
| Random state   | 42             |
| Stratification | Yes            |
| Testing rows   | 1,652          |
| Features       | 10             |

## 5. Exploratory Data Analysis

The notebook performs exploratory analysis to understand transaction behavior, customer segments, product categories, and the distribution of the next-purchase target.

### EDA Areas Covered

- Customer segment distribution
- Product category distribution
- Monthly transaction patterns
- Sales and transaction behavior
- Purchase frequency
- Next-purchase category distribution

### Why EDA Matters

**EDA helps identify customer behavior patterns and provides context before converting historical transactions into predictive customer-level features.**

### Modeling Note

The notebook's measured results show that this is a challenging multi-class prediction problem: the final XGBoost Top-1 accuracy is 11.74%, while Top-5 accuracy reaches 49.76%.

## 6. Feature Engineering

Customer-level behavioral features are generated from historical transactions. These features summarize purchase frequency, spending, product diversity, returns, and recency.

| Feature                  | What it represents                      |
|--------------------------|-----------------------------------------|
| total_orders             | Number of historical orders             |
| total_spend              | Total customer spending                 |
| avg_order_value          | Average order value                     |
| avg_quantity             | Average quantity purchased              |
| avg_discount             | Average discount percentage             |
| avg_delivery_days        | Average delivery time                   |
| unique_products          | Number of unique products purchased     |
| unique_categories        | Number of unique categories purchased   |
| return_rate              | Share of returned transactions          |
| days_since_last_purchase | Recency of the customer's last purchase |

**These ten features form the main behavioral input used by the classification models.**



## 7. Machine Learning Models

The project evaluates a baseline classifier and three Machine Learning algorithms for the multi-class next-purchase prediction task.

| Model               | Role                               | Notebook Accuracy |
|---------------------|------------------------------------|-------------------|
| Baseline            | Most-common-category benchmark     | 11.86%            |
| Logistic Regression | Linear classification baseline     | 12.71%            |
| Random Forest       | Tree-based ensemble                | 10.11%            |
| XGBoost             | Gradient-boosted multi-class model | 11.74%            |

### XGBoost Configuration

```
n_estimators = 300
max_depth = 6
learning_rate = 0.05
subsample = 0.8
colsample_bytree = 0.8
eval_metric = mlogloss
```

### Selected Prediction Workflow

Although Logistic Regression records the highest Top-1 accuracy in the notebook, the project uses XGBoost in the prediction and recommendation workflow because the deployment application is built around its probability outputs and Top-5 ranking.

## 8. Model Comparison & Evaluation

### Accuracy Comparison

| Rank | Model               | Accuracy |
|------|---------------------|----------|
| 1    | Logistic Regression | 12.71%   |
| 2    | Baseline            | 11.86%   |
| 3    | XGBoost             | 11.74%   |
| 4    | Random Forest       | 10.11%   |

### XGBoost Classification Report

| Metric            | Value |
|-------------------|-------|
| Accuracy          | 0.12  |
| Macro F1          | 0.07  |
| Weighted F1       | 0.10  |
| Test observations | 1,652 |

**Important: the notebook's direct Top-1 accuracy is modest because the task contains many purchase categories. The project therefore also evaluates ranked Top-K predictions.**

## 9. Top-K Prediction & Recommendation

### Top-K Evaluation

| Prediction Scope | Accuracy |
|------------------|----------|
| Top-1            | 11.74%   |
| Top-3            | 32.75%   |
| Top-5            | 49.76%   |

The Top-K approach returns several likely categories rather than forcing the system to rely on only one class. This is especially useful for recommendation-oriented applications.

### Example Prediction from the Notebook

| Rank | Category        | Probability |
|------|-----------------|-------------|
| 1    | Grocery         | 58.00%      |
| 2    | Computers       | 7.79%       |
| 3    | Bags            | 4.93%       |
| 4    | Home Appliances | 4.86%       |
| 5    | Fashion         | 4.24%       |

### Personalized Product Recommendation

The recommendation function takes the top predicted category and selects products from that category, ranking them by product rating. The notebook's example produces product-level recommendations such as Amul Staples 0035 and Nestle Snacks products for a Grocery prediction.

## 10. Streamlit Application

The project includes a Streamlit application named Customer Next Purchase AI. Users select a customer and generate an AI-powered prediction.

### Application Flow

| Step | Application action                |
|------|-----------------------------------|
| 1    | Select Customer                   |
| 2    | View Customer Overview            |
| 3    | Review Purchase History           |
| 4    | Generate Next Purchase Prediction |
| 5    | View Top-5 Probabilities          |
| 6    | Receive Personalized Products     |

### Application Displays

- Customer segment and location
- Total orders
- Total spend
- Average order value
- Days since last purchase
- Purchase history
- Most likely next purchase category
- Top-5 prediction probabilities
- Personalized product recommendations

**Deployment model: XGBoost + label encoder + customer/product/transaction data → Top-5 prediction → personalized recommendation.**

# 11. Project Structure

Recommended GitHub folder structure for the second project:

```
02-Customer-Next-Purchase-Prediction/  
├── README.md  
├── customer_next_purchase_prediction.ipynb  
├── deployment.ipynb  
├── app.py  
├── customer_next_purchase_dataset.xlsx  
├── next_purchase_model.pkl  
└── label_encoder.pkl
```

## Purpose of Each File

| File                                    | Purpose                                |
|-----------------------------------------|----------------------------------------|
| README.md                               | Interactive project documentation      |
| customer_next_purchase_prediction.ipynb | Complete data analysis and ML workflow |
| deployment.ipynb                        | Deployment preparation                 |
| app.py                                  | Streamlit application                  |
| customer_next_purchase_dataset.xlsx     | Customers, products and transactions   |
| next_purchase_model.pkl                 | Saved trained model                    |
| label_encoder.pkl                       | Saved category encoder                 |

Note: the notebook and application currently reference local development paths. For GitHub deployment, project-relative paths should be used.

## 12. Key Takeaways & Future Improvements

### Key Takeaways

- Converted chronological purchase history into a supervised next-purchase prediction problem.
- Created ten customer-level behavioral features.
- Compared baseline, Logistic Regression, Random Forest, and XGBoost.
- Evaluated Top-1, Top-3 and Top-5 prediction performance.
- Extended category prediction into personalized product recommendations.
- Integrated the workflow into an interactive Streamlit application.

### Future Improvements

- Sequential and time-aware recommendation models
- Hyperparameter optimization and class-balancing strategies
- More behavioral and temporal features
- Recommendation ranking models
- Collaborative filtering or hybrid recommendation systems
- Real-time customer data
- Automated model retraining
- Cloud deployment and monitoring

**Project Direction: Analyze → Learn → Predict → Recommend → Deploy**

Thank you for exploring this project!

Source: Customer Next Purchase Prediction Jupyter Notebook and accompanying Streamlit deployment application supplied with this project.